COMMERCIAL SAAS ARCHITECTURE

EduManage Cloud Platform

Complete Business Model, Revenue Architecture & Commercial Strategy

Platform Architecture: ASP.NET Core 8 Web API • Angular 18/19 SPA • SQL Server • EF Core 8
Scope: Multi-Tenant SaaS Provider Monetization & Institutional Fee Billing Operations

1. Executive Summary & Market Positioning

EduManage Cloud is engineered as a high-margin, multi-tenant B2B SaaS platform for the modern educational ecosystem. It unifies primary/secondary schools (K-12) and collegiate degree institutions within a single modular code-base, eliminating fragmented software purchases.

The platform deploys a **Dual-Engine Financial Architecture**:

> **Dual Financial Architecture:**
> • **Engine A (SaaS Provider Level):** Predictable recurring subscription revenue (MRR & ARR) collected from school trusts and college groups.
> • **Engine B (Institution Operations Level):** Granular student fee scheduling, scholarship discounting, batch invoice generation, multi-mode collections (Cash/Bank/UPI), and overdue balance aging.

2. Engine A: Platform SaaS Subscription Business Model (B2B)

Educational institutions pay the SaaS provider on a monthly or annual subscription basis. Three targeted commercial tiers match institutional size and growth:

Subscription Metric	Starter Tier	Standard Tier (Recommended)	Enterprise Tier
Monthly Pricing	₹1,999 / month	**₹4,999 / month**	₹9,999 / month
Annual Pricing	₹19,990 / year (Save 17%)	**₹49,990 / year (Save 17%)**	₹99,990 / year (Save 17%)
Target Audience	K-10 primary schools, coaching institutes	**K-12 schools, junior degree colleges**	Multi-campus universities & trusts
Enrolled Student Limit	Up to 300 students	**Up to 1,500 students**	Up to 10,000+ students
Campuses Allowed	1 Campus	**Up to 3 Campuses**	Unlimited Campuses
Academic Scope	School grades, classes & sections	**School classes + College degree programs**	Full university hierarchy & credits

Subscription Metric	Starter Tier	Standard Tier (Recommended)	Enterprise Tier
Payment Collection	Manual cash, cheque & bank entry	**Integrated Online Gateway + Manual**	Custom gateway split & accounts
Auditing & SLA	Standard support	**Automated billing background reminders**	Full audit log trail & dedicated SLA

SaaS Revenue Metrics & Telemetry

The built-in SaaS Provider Telemetry dashboard calculates:

- **MRR (Monthly Recurring Revenue):** Total recurring fees recognized on a monthly cadence across all active tenant institutions.
- **ARR (Annual Recurring Revenue):** Projected annual contract value (`MRR × 12`).
- **Seat Expansion Rate:** Automatic upsell trigger when a tenant approaches 90% of their enrolled student limit.

3. Engine B: Institutional Fee & Revenue Billing Model (B2C)

Each subscribed institution operates an independent, compliant financial ledger to manage student admissions, tuition, custom scholarship concessions, installment plans, and digital collections:

A. Modular Line-Item Fee Structures

Institutions configure itemized fees rather than opaque lump sums: Tuition fee, Laboratory fee, Library & Activity fee, Sports fee, Examination fee, and Route-based Transportation fee.

B. Flexible Billing Frequencies

Supports Monthly (preschools/coaching), Quarterly/Term-wise (private K-12), Semester-wise (colleges), and Annual upfront payment cycles.

Scholarships & Student-Specific Concessions

Institutions can assign customized fee assignments to individual students:

- **Percentage Discounts:** e.g., 10% to 50% concession for merit scholars or sports achievers.
- **Fixed Flat Concessions:** e.g., Flat ₹5,000 concession with explicit audit reason recorded in the database.
- **Net Invoicing Formula:** `Invoice Total = (Sum of Line Items) - Discount Amount`.

Multi-Channel Payment Collection Engine

The platform supports both traditional offline collections and modern digital payments:

- **Manual Cash, Cheque & Bank Transfers:** Front-desk accountants record payments directly, generating sequential receipts (`RCT-2026-0001`).
- **Online Payment Gateway Integration:** Supports instant online checkout via Razorpay and Stripe (UPI, QR Code, Debit/Credit Cards, Net Banking).
- **Webhook Deduplication & Idempotency:** Online payments are only marked successful via verified cryptographic webhook events, eliminating browser redirect spoofing.
- **Partial Payments & Status Progression:** Invoices dynamically transition across: `Issued` → `PartiallyPaid` → `Paid` → `Overdue`.

4. Dual Academic Delivery Models

Institutions are not forced into rigid structures. EduManage provides native data models for both education formats:

Dimension	K-12 School Model	Higher Education / College Model
Organizational Unit	Academic Year → Grade/Class (e.g., Grade 10)	Department → Degree Program (e.g., B.Tech, B.Sc)
Student Cohort	Section (e.g., Section A, Section B)	Semester Cohort (Semesters 1 through 8)
Classroom Supervision	Dedicated Class Teacher assigned per section	Head of Department (HOD) + Course Offering Faculty

Dimension	K-12 School Model	Higher Education / College Model
Curriculum Structure	Core school subjects (Math, Science, English)	Course offerings with Credit units & Elective tags
Grading & Transcripts	Continuous unit tests, Mid-Term, Final exams	Semester assessments, GPA grading, Transcript cards

5. Technical Monetization & Infrastructure Cost Analysis

To ensure exceptional operating margins, the system uses a **modular monolith architecture** that completely eliminates expensive microservice networking and unneeded enterprise server overhead:

Infrastructure Component	Recommended Low-Cost Provider	Monthly Cost (INR)
Frontend Static Hosting	Cloudflare Pages / Vercel (Global Edge CDN, Unlimited SSL)	**₹0 (Free Tier)**
Backend API (.NET 8)	Hetzner Cloud CPX11 / DigitalOcean (2 vCPU, 2GB RAM, Ubuntu)	~₹450 / month (€4.50)
Managed Database (SQL Server)	Azure SQL Database (Basic 5 DTU, 2GB) or VPS SQL instance	~₹420 / month ($4.99)
Domain, DNS & DDoS Defense	Cloudflare DNS + Standard .com / .in domain	~₹75 / month (₹900/year)
Transactional Email (Notifications)	Brevo / Resend Free Tier (300 emails/day)	₹0 (Free Tier)
Total Monthly Infrastructure Cost	**Complete Production Footprint**	**~₹945 – ₹1,200 / month**

> **Profit Margin Analysis:**
> With just **1 institution on the Standard Plan (₹4,999/mo)**, the platform generates **~75% gross profit margin** over total hosting costs. At **10 institutions**, monthly gross profit exceeds **97%**.

6. Multi-Tenant Data Isolation & Security Governance

Data security is guaranteed at the architectural level to ensure institutional privacy and regulatory compliance:

- **Row-Level Security (EF Core Global Filters):** All queries execute with an automatic `WHERE TenantId = @CurrentTenant` predicate, preventing any cross-school data visibility.
- **Cryptographic JWT Claims:** The tenant context is resolved exclusively from verified JWT token claims; user-supplied headers or URL query parameters cannot manipulate tenant access.
- **Audit Trails:** Financial allocations, fee waivers, grade revisions, and login events are permanently recorded with timestamps, user IDs, and client IP addresses.

7. Strategic Expansion & Add-On Monetization Roadmap

Additional revenue streams can be unlocked through optional platform modules:

1. **Payment Gateway Convenience Fee Share:** Earn a 0.20% – 0.50% margin on digital fee transaction volumes processed through platform-integrated Razorpay/Stripe merchant accounts.

2. **SMS & WhatsApp Parent Notification Packs:** Sell bundled communication credits for automated fee reminders and attendance circulars.
3. **Hardware Attendance Device Add-On:** Supply pre-configured RFID and biometric fingerprint readers with cloud webhook syncing.
4. **Branded Mobile Apps:** White-label native iOS and Android apps for schools and colleges for an annual add-on fee.
